

Local projections versus VAR

Status: draft for review

Question and intuition

Local projections estimate a separate regression for each horizon rather than repeatedly applying a single fitted transition equation. This permits horizon-specific dynamics but can increase sampling variance. Neither method supplies identification by itself: the shock and control assumptions must still be credible.

Model and assumptions

$y_{t+h} = a_h + \theta_h z_t + \Gamma_h X_t + e_{t+h}$. Here z_t is an identified policy innovation in rate units. Overlapping dependent outcomes induce horizon-dependent residual dependence. Newey-West HAC with a reported bandwidth estimates the long-run covariance of the regression score.

Interpretation and common errors

A VAR borrows structure across horizons. LP trades those restrictions for flexibility. In small monthly samples, high-dimensional controls can exhaust the degrees of freedom at long horizons; the correct result is unavailable inference, not extrapolation. An LP using a point-estimated VAR shock is conditional on that shock; a joint bootstrap can propagate shock-estimation uncertainty, while ordinary HAC does not.

Portfolio application

For cumulative inflation, construct the cumulative dependent outcome before fitting the regression and estimating HAC. Summing standard errors across horizons is invalid because their estimates are correlated. Pre-shock price levels enter as nuisance controls. The pass-through LP is normalized to a TRY depreciation normalized to 100 times $\log(1.10)$, exactly a ten percent increase in TRY per USD. The recent TR regime has only four domestic variables and one lag, with unavailable long horizons left missing.

Exam practice

1. Why estimate each horizon separately?

Answer. To avoid imposing one VAR transition equation on all response horizons.

2. Why use HAC?

Answer. Overlapping outcomes and serial dependence correlate the regression scores.

3. Can LP fix an invalid policy shock?

Answer. No. Identification remains a separate assumption.

4. How should cumulative-response uncertainty be computed?

Answer. Estimate a cumulative dependent outcome directly or propagate the full covariance jointly.

5. Why omit a long-horizon estimate in a short regime?

Answer. The remaining sample may lack residual degrees of freedom.

Worked reasoning

Before interpreting an estimate, name the outcome unit, shock unit, identifying restrictions, sample and uncertainty method. If any is missing, the numerical sign alone does not support a causal claim. Separate a mechanical accounting identity from a maintained economic restriction. Explain how the answer would change under a different permissible identification without searching until a preferred answer appears. A correct exam response states what can and cannot be inferred, rather than merely naming a test or estimator.

Reference and further reading

Verified primary reference: <https://doi.org/10.1257/0002828053828518>

These notes are original explanations written for this portfolio; no thesis text or code was used.